FlashSpace Detailed Database Schema Documentation

This document provides a comprehensive technical reference of all primary database schemas within the FlashSpace ecosystem. FlashSpace uses **MongoDB** with **Typegoose** for data modeling.

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1. User Management

`User` Schema

The primary entity representing individuals, partners, and administrators.

Field Name	Type	Description
`email`	String	Unique identifier for authentication. Lowercased and trimmed.
`fullName`	String	Full name of the user.
`phoneNumber`	String	User's mobile number (optional).
`profilePicture`	String	URL to the user's avatar image.
`password`	String	Hashed password (hidden by default).
`googleId`	String	Unique ID for Google OAuth users.
`authProvider`	Enum	`local` (email/password) or `google`.
`role`	Enum	Permissions level: `user`, `admin`, `super_admin`, `partner`, `sales`, `affiliate`, etc.
`isEmailVerified`	Boolean	True if the email has been validated via OTP/Token.
`kycVerified`	Boolean	True if the user's KYC has been fully approved.
`isActive`	Boolean	Determines if the account is allowed to log in (Default: `true`).
`isDeleted`	Boolean	Soft-deletion flag.
`credits`	Number	Wallet balance for the user (Default: `0`).
`lastLogin`	Date	Timestamp of the most recent successful authentication.
`preferences`	Object	Language, Currency (`inr` by default), `darkMode`, `timeZone`, etc.
`notifications`	Object	Toggles for `email`, `push`, `promotional`, `reminders`, and `loginAlerts`.
`security`	Object	Preferences for `sessionManagement` and `dataSharing`.
`twoFactor`	Boolean	Status of Two-Factor Authentication.

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2. Property & Infrastructure

`Property` Schema

The parent container for physical spaces.

Field Name	Type	Description
`name`	String	Commercial name of the building/complex.
`address`	String	Full street address.
`city`	String	City location (indexed for search).
`area`	String	Specific locality within the city (indexed for search).
`features`	String[]	Tag-based list of facilities (e.g., "Cafe", "Gym", "Parking").
`location`	GeoLocation	GeoJSON Point containing `coordinates` [long, lat].
`images`	String[]	Array of URLs for property photos.
`kycStatus`	Enum	`approved`, `rejected`, `pending`, `not_started`.
`kycRejection`	String	Detail on why a property's KYC was denied.
`documents`	Item[]	List of uploaded property-specific legal documents.
`status`	Enum	Property lifecycle: `draft`, `pending_approval`, `active`, `suspended`.
`isActive`	Boolean	Toggle to display/hide property on the frontend.
`partner`	Ref(User)	Reference to the Partner account managing this property.

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3. Product Offerings (Spaces)

`CoworkingSpace` Schema

Desks and hot-seating modules.

Field Name	Type	Description
`property`	Ref(Prop)	Reference to the parent Building.
`capacity`	Number	Total number of seats available.
`partnerPrice`	Number	Monthly base rate set by the partner.
`adminMarkup`	Number	Commission added by FlashSpace.
`finalPrice`	Number	Aggregate price shown to the customer.
`floors`	Floor[]	Nested hierarchy: Floor -> Table -> Seat (for precise booking).
`operatingHours`	Object	Schedule of operations (Open/Close times per day).
`avgRating`	Number	Calculated average from all customer reviews.
`totalReviews`	Number	Count of reviews received.
`amenities`	String[]	Space-specific perks (e.g., "Ergonomic Chairs", "Coffee").
`images`	String[]	Photos specific to the coworking area.

`approvalStatus`	Enum	Verification state of the space.
`sponsored`	Boolean	Highlight in search results (paid promotion).
`popular`	Boolean	High-demand badge.

`MeetingRoom` Schema

Hourly booking units for meetings/conferences.

Field Name	Type	Description
`type`	Enum	`meeting_room`, `board_room`, `conference_room`.
`capacity`	Number	Maximum occupancy.
`count`	Number	Quantity of identical rooms available.
`partnerPriceH`	Number	Hourly rate for the partner.
`finalPriceH`	Number	Hourly rate for the client.
`partnerPriceD`	Number	Optional daily rate for the partner.
`minBookingHrs`	Number	Minimum time slot required for a booking.
`operatingHours`	Object	Availability schedule.
`amenities`	String[]	Equipment like "Projector", "Whiteboard", "VC System".

`VirtualOffice` Schema

Address services for business registration.

Field Name	Type	Description
`finalGstPrice`	Number	Annual cost for GST Address plan.
`finalMailPrice`	Number	Annual cost for Mailing Address plan.
`finalBrPrice`	Number	Annual cost for Business Registration plan.
`amenities`	String[]	Included services (e.g., "Mail Forwarding", "Call Handling").
`approvalStatus`	Enum	Internal status of the VO listing.

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4. Transactions & Operations

`Booking` Schema

Records of finalized sales.

Field Name	Type	Description
`bookingNumber`	String	Unique Invoice/Booking ID (e.g., BK-XXXX).

`user`	Ref(User)	Client who made the booking.
`partner`	Ref(User)	Space provider fulfilling the booking.
`type`	Enum	Type of space booked (`VirtualOffice`, `CoworkingSpace`, etc.).
`spaceId`	Ref	Dynamic reference to the specific space item.
`spaceSnapshot`	Object	Metadata (Name, Address, City) cached at time of purchase.
`plan`	Object	Details of the purchased plan (Price, Tenure, Units).
`status`	Enum	`pending_payment`, `pending_kyc`, `active`, `expired`, `cancelled`.
`kycStatus`	Enum	State of documents submitted for *this specific* booking.
`timeline`	Entry[]	Audit trail of status changes (Status, Date, Note).
`documents`	Item[]	Generated licenses, invoices, or agreements.
`startDate`	Date	Activation date of the service.
`endDate`	Date	Expiration date.
`couponCode`	String	Applied discount identifier.

`Payment` Schema

Financial tracking via Razorpay gateway.

Field Name	Type	Description
`razorpayOrderId`	String	Unique ID from Payment Gateway.
`amount`	Number	Transaction amount in Paisa (100 = Rs. 1.00).
`status`	Enum	`pending`, `processing`, `completed`, `failed`, `refunded`.
`paymentType`	Enum	Categorization based on product type.
`spaceName`	String	Name of the space being paid for.
`planName`	String	Selected plan duration/type.
`tenure`	Number	Duration in years/months.
`totalAmount`	Number	Grand total inclusive of all fees.
`creditsUsed`	Number	Amount deducted from user's credit balance.
`refundId`	String	Gateway identifier for successful refunds.

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5. Support & Engagement

`KYCDocument` Schema

Compliance repository for Users and Partners.

Field Name	Type	Description
`profileName`	String	Label for the profile (e.g., "Company A - GST").
`kycType`	Enum	`individual` or `business`.
`personalInfo`	Object	`aadhaar`, `pan`, `dob`, `phone`.
`businessInfo`	Object	`gstNumber`, `companyType`, `registeredAddress`.
`documents`	Item[]	Detailed list: `type`, `fileUrl`, `verifiedAt`, `rejectionReason`.
`kycStatus`	Enum	`approved`, `rejected`, `resubmit`, `pending`.
`progress`	Number	Percentage completion of the profile.

`SupportTicket` Schema

Customer service management.

Field Name	Type	Description
`ticketNumber`	String	Auto-generated support ID.
`subject`	String	Summary of the issue.
`category`	Enum	`billing`, `kyc`, `technical`, `booking`, etc.
`priority`	Enum	`low`, `medium`, `high`, `urgent`.
`status`	Enum	`open`, `in_progress`, `resolved`, `closed`.
`messages`	Msg[]	Thread of messages between user and support.
`assignedTo`	Ref(User)	Admin/Support staff handling the ticket.

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6. Shared Structural Schemas

`GeoLocation`

```
{
  type: "Point",
  coordinates: [longitude, latitude] // [Number, Number]
}
```

`OperatingHours`

```
{
  monday:    { open: string, close: string, isOpen: boolean },
  tuesday:   { open: string, close: string, isOpen: boolean },
  // ... and so on for all 7 days
}
```

7. Entity Relationship Model

```
erDiagram
    USER ||--o{ PROPERTY : "owns/manages"
    USER ||--o{ BOOKING : "buys"
    USER ||--o{ PAYMENT : "makes"
    USER ||--o{ KYCDocument : "owns"
    USER ||--o{ SUPPORT_TICKET : "files"
    PROPERTY ||--o{ COWORKING_SPACE : "hosts"
    PROPERTY ||--o{ MEETING_ROOM : "hosts"
    PROPERTY ||--o{ VIRTUAL_OFFICE : "hosts"
    COWORKING_SPACE ||--o{ SEAT_BOOKING : "tracks"
    BOOKING ||--o| PAYMENT : "linked"
    BOOKING ||--o| KYCDocument : "requires"
```